

Foxconn Open BMC 新人訓

L3 - Multi Function Pin and DTS

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環境設定

系統資訊

項目	內容
作業系統	Ubuntu 22.04.5 LTS (VirtualBox)
核心版本	6.8.0-87-generic
Python	3.10.12
硬碟空間	1TB (約 750GB 可用)

OpenBMC 環境

項目	內容
OpenBMC 版本	2.18.0 (commit: 05f7a3f2ed)
MACHINE	romulus
原始碼路徑	~/openbmc
Build 路徑	~/openbmc/build/romulus
映像檔路徑	~/openbmc/build/romulus/tmp/deploy/images/romulus/

作業目標

本次作業目標是學習 Linux Device Tree Source (DTS) 的配置方式, 透過修改 Kernel DTS 來停用 I2C Bus, 實際體驗硬體配置與軟體的關聯。

項目	說明
原始作業要求	Disable I2C15 in U-Boot and Kernel DTS
實際執行目標	Disable I2C12 in Kernel DTS
調整原因	romulus 使用 AST2500 晶片, 僅支援 i2c0~i2c13, 無 i2c15

背景知識

Aspeed 晶片 I2C 架構

AST2500 晶片內建 14 個 I2C Controller, 每個 Controller 對應一條 I2C Bus:

項目	說明
I2C Controller	SoC 內部硬體模組, 負責產生 I2C 訊號
I2C Bus	每條 Bus 僅有 2 根 Pin(SDA 資料線 + SCL 時脈線)
I2C Device	掛在 Bus 上的設備, 透過 7-bit 地址區分(0x00~0x7F)
記憶體位址	CPU 存取 Controller 暫存器的位址(如 i2c12 = 0x1e78a440)

I2C Bus 連接方式

一條 I2C Bus 上可以掛載多個設備, 所有設備並聯共用 SDA 和 SCL 兩條線, 透過不同的 I2C 地址來區分設備。

Device Tree Source (DTS)

DTS 是描述硬體結構的文字檔案, 用於告訴 Linux Kernel 硬體的配置方式。透過修改 DTS 中的 status 屬性, 可以啟用或停用硬體功能:

- status = "okay" → 啟用該硬體
- status = "disabled" → 停用該硬體

實作步驟

步驟 1: 確認開發環境

目的: 進入 OpenBMC build 環境, 準備進行 Kernel 修改。

```
cd ~/openbmc
source setup romulus
```

步驟 2: 分析 romulus 的 I2C 配置

目的: 確認 romulus 板實際啟用哪些 I2C Bus, 以及 i2c12 上掛載的設備。

```
grep -A3 "&i2c"
~/openbmc/build/romulus/tmp/work-shared/romulus/kernel-source/arch/arm/boot/d
ts/aspeed/aspeed-bmc-opp-romulus.dts
```

分析結果:

項目	內容
晶片型號	AST2500
硬體支援範圍	i2c0 ~ i2c13 (共 14 個 Controller)
romulus 實際啟用	i2c1 ~ i2c12 (共 12 個, i2c0 和 i2c13 未連接設備)
i2c12 掛載設備	w83773g@4c (溫度感測器, QEMU 未模擬此設備)

步驟 3: 驗證修改前狀態

目的: 確認 i2c12 在修改前是存在的, 作為後續比較的基準。

啟動 QEMU 並 SSH 進入, 執行以下指令:

```
i2cdetect -l
i2cdetect -y 12
```

測試結果:

```
sonny@sonny-VirtualBox: ~/openbmc/build/romulus
[ OK ] Started Phosphor Settings Daemon.
[ OK ] Started Avahi mDNS/DNS-SD Stack.
[ OK ] Started OpenBMC Software Update Manager.
[ OK ] Started Phosphor Dump Manager.
[ OK ] Started Phosphor Power0 Control.
[ OK ] Started Phosphor Button Handler.
[ OK ] Started Phosphor systemd target monitor.
[ OK ] Started Sensor Monitor.
[ OK ] Started Serial Getty on ttyS4.
[ OK ] Reached target Login Prompts.
Terminal Starting Phosphor Sysfs - Add LED...
Starting Phosphor Sysfs - Add LED...
Starting Phosphor Sysfs - Add LED...
Starting Hostname Service...
Starting Phosphor LED Group Management Daemon...
Starting Phosphor Ldap config updater...
Starting Wait for /org/openbmc/control/power0...
Starting Wait for /xyz/openbmc-project/control/host0/auto_reboot...
Starting Wait for /xyz/openbmc-project/control/host0/boot/one_time...
Starting Wait for /xyz/openbmc-project/control/host0/boot...
Starting Wait for /xyz/openbmc-project/control/host0/power_restore_policy...
Starting Wait for /xyz/openbmc-project/control/host0/restriction_node...
Starting Wait for /xyz/openbmc-project/state/bmc0...
Starting Wait for /xyz/openbmc-project/state/host0...
Starting Wait for /xyz/openbmc-project/ttne/sync_method...
[ OK ] Started Phosphor Hwmon Poller.
[ OK ] Started Phosphor Hwmon Poller.
Starting Phosphor Version Software Manager...
Starting Phosphor Chassis0 State Manager...
[ OK ] Started Hostname Service.
[ OK ] Finished Phosphor Sysfs - Add LED.
[ OK ] Finished Phosphor Sysfs - Add LED.
[ OK ] Finished Phosphor Sysfs - Add LED.
[ OK ] Finished Wait for /org/openbmc/control/power0.
[ OK ] Finished Wait for /xyz/openbmc-project/control/host0/auto_reboot.
[ OK ] Finished Wait for /xyz/openbmc-project/control/host0/boot/one_time.
[ OK ] Finished Wait for /xyz/openbmc-project/control/host0/boot.
[ OK ] Finished Wait for /xyz/openbmc-project/control/host0/power_restore_policy.
[ OK ] Finished Wait for /xyz/openbmc-project/control/host0/restriction_node.
[ OK ] Finished Wait for /xyz/openbmc-project/ttne/sync_method.
Phosphor OpenBMC (Phosphor OpenBMC Project Reference Distro) nodistro.0 romulus ttyS4
romulus login: qemu-system-arm -n 256 -M romulus-bmc -nographic \
Phosphor OpenBMC (Phosphor OpenBMC Project Reference Distro) nodistro.0 romulus ttyS4
romulus login:

sonny@sonny-VirtualBox: ~
sonny@sonny-VirtualBox: $ ssh -p 2222 root@127.0.0.1
root@127.0.0.1's password:
root@romulus:~# i2cdetect -l
i2c-1 i2c 1e78a080.i2c I2C adapter
i2c-2 i2c 1e78a0c0.i2c I2C adapter
i2c-3 i2c 1e78a100.i2c I2C adapter
i2c-4 i2c 1e78a140.i2c I2C adapter
i2c-5 i2c 1e78a180.i2c I2C adapter
i2c-6 i2c 1e78a1c0.i2c I2C adapter
i2c-7 i2c 1e78a300.i2c I2C adapter
i2c-8 i2c 1e78a340.i2c I2C adapter
i2c-9 i2c 1e78a380.i2c I2C adapter
i2c-10 i2c 1e78a3c0.i2c I2C adapter
i2c-11 i2c 1e78a400.i2c I2C adapter
i2c-12 i2c 1e78a440.i2c I2C adapter
root@romulus:~# i2cdetect -y 12
0 1 2 3 4 5 6 7 8 9 a b c d e f
00: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
10: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
60: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
70: -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
root@romulus:~#
```

修改前 i2c-12 存在於系統中, i2cdetect -y 12 可正常執行掃描(全部顯示 -- 是因為 QEMU 沒有模擬實體 I2C 設備)。

步驟 4: 使用 devtool 修改 Kernel

目的: 使用 devtool 建立隔離的開發環境, 避免修改被 bitbake 覆蓋。

devtool modify linux-aspeed

執行後, Kernel 原始碼會複製到:

~/openbmc/build/romulus/workspace/sources/linux-aspeed/

步驟 5: 備份並修改 DTS 檔案

目的: 修改 i2c12 的 status 屬性, 將其停用。

備份原始檔案:

```
cp ./arch/arm/boot/dts/aspeed/aspeed-bmc-opp-romulus.dts
~/Newcomer_training/L3/aspeed-bmc-opp-romulus.dts.orig
```

修改 DTS(使用 sed 指令):

```
sed -i '225s/okay/disabled/'
./arch/arm/boot/dts/aspeed/aspeed-bmc-opp-romulus.dts
```

修改前後對比:

```

sonny@sonny-VirtualBox:~/openbmc/build/romulus/workspace/sources/linux-aspeed$ ls -la ~/Newcomer_training/L3/
total 24
drwxrwxr-x  2 sonny sonny 4096 十二  3 15:51 .
drwxrwxr-x 16 sonny sonny 4096 十一 28 14:48 ..
-rw-r--r--  1 sonny sonny 6257 十二  3 15:51 aspeed-bmc-opp-romulus.dts.modified
-rw-r--r--  1 sonny sonny 6253 十二  3 15:41 aspeed-bmc-opp-romulus.dts.orig
sonny@sonny-VirtualBox:~/openbmc/build/romulus/workspace/sources/linux-aspeed$ diff ~/Newcomer_training/L3/aspeed-bmc-opp-romulus.dts.orig ~/Newcomer_training/L3/aspeed-bmc-opp-romulus.dts.modified
225c225
<     status = "okay";
---
>     status = "disabled";

```

行號	修改前	修改後
225	status = "okay";	status = "disabled";

步驟 6: 編譯修改後的 Image

目的: 編譯包含修改的 Kernel, 並重新打包 BMC Image。

```
cd ~/openbmc/build/romulus
devtool build linux-aspeed
bitbake obmc-phosphor-image
```

步驟 7: 驗證修改後狀態

目的: 確認 i2c12 已被成功停用。

關閉舊 QEMU, 使用新編譯的 Image 重新啟動, SSH 進入測試:

```
i2cdetect -l
i2cdetect -y 12
```

測試結果:

```
sonny@sonny-VirtualBox: ~/openbmc/build/romulus
[ OK ] Started Phosphor Buttons.
[ OK ] Started Telemetry.
[ OK ] Started Phosphor Download Manager.
[ OK ] Started OpenPower Software Update Manager.
[ OK ] Started Phosphor Inventory Manager.
[ OK ] Started OpenBMC Software Update Manager.
[ OK ] Started Avahi mDNS/DNS-SD Stack.
[ OK ] Started Phosphor Power0 Control.
[ OK ] Started Phosphor Dump Manager.
[ OK ] Started Phosphor Button Handler.
[ OK ] Started phosphor systemd target monitor.
[ OK ] Started Sensor Monitor.
[ OK ] Started Serial Getty on ttyS4.
[ OK ] Reached target Login Prompts.
Starting Phosphor Sysfs - Add LED...
Starting Phosphor Sysfs - Add LED...
Starting Phosphor Sysfs - Add LED...
Starting Hostname Service...
Starting Phosphor LED Group Management Daemon...
Starting Phosphor Imap config updater...
Starting Wait for /org/openbmc/control/power0...
Starting Wait for /xyz/openbmc_project/control/host0/auto_reboot...
Starting Wait for /xyz/openbmc_project/control/host0/boot/one_time...
Starting Wait for /xyz/openbmc_project/control/host0/power_restore_policy...
Starting Wait for /xyz/openbmc_project/control/host0/restriction_mode...
Starting Wait for /xyz/openbmc_project/state/bmc0...
Starting Wait for /xyz/openbmc_project/state/host0...
Starting Wait for /xyz/openbmc_project/tline/sync_method...
[ OK ] Started Phosphor Hwmon Poller.
[ OK ] Started Phosphor Humon Poller.
Starting Phosphor Version Software Manager...
Starting Phosphor Chassis0 State Manager...
Starting Hostname Service.
[ OK ] Finished Phosphor Sysfs - Add LED.
[ OK ] Finished Phosphor Sysfs - Add LED.
[ OK ] Finished Phosphor Sysfs - Add LED.
[ OK ] Finished Wait for /org/openbmc/control/power0.
[ OK ] Finished Wait for /xyz/openbmc_project/control/host0/auto_reboot.
[ OK ] Finished Wait for /xyz/openbmc_project/control/host0/boot/one_time.
[ OK ] Finished Wait for /xyz/openbmc_project/control/host0/power_restore_policy.
[ OK ] Finished Wait for /xyz/openbmc_project/control/host0/restriction_mode.
[ OK ] Finished Wait for /xyz/openbmc_project/tline/sync_method.
Phosphor OpenBMC (Phosphor OpenBMC Project Reference Distro) nodistro.0 romulus ttyS4
romulus login:

sonny@sonny-VirtualBox: ~/openbmc/build/romulus
root@romulus:~# i2cdetect -l
i2c-1 i2c 1e78a080.i2c I2C adapter
i2c-2 i2c 1e78a0c0.i2c I2C adapter
i2c-3 i2c 1e78a100.i2c I2C adapter
i2c-4 i2c 1e78a140.i2c I2C adapter
i2c-5 i2c 1e78a180.i2c I2C adapter
i2c-6 i2c 1e78a1c0.i2c I2C adapter
i2c-7 i2c 1e78a300.i2c I2C adapter
i2c-8 i2c 1e78a340.i2c I2C adapter
i2c-9 i2c 1e78a380.i2c I2C adapter
i2c-10 i2c 1e78a3c0.i2c I2C adapter
i2c-11 i2c 1e78a400.i2c I2C adapter
root@romulus:~# i2cdetect -y 12
Error: Could not open file '/dev/i2c-12' or '/dev/i2c/12': No such file or directory
root@romulus:~#
```

修改後 i2c-12 不存在於系統中, i2cdetect -y 12 顯示錯誤:

```
Error: Could not open file '/dev/i2c-12' or '/dev/i2c/12': No such file or directory
```

步驟 8: 還原開發環境

目的: 測試完成後, 還原乾淨的開發環境。

```
devtool reset linux-aspeed
bitbake obmc-phosphor-image
```

```
sonny@sonny-VirtualBox:~/openbmc/build/romulus$ cd ~/openbmc/build/romulus
sonny@sonny-VirtualBox:~/openbmc/build/romulus$ devtool reset linux-aspeed
NOTE: Starting bitbake server...
INFO: Cleaning sysroot for recipe linux-aspeed...
INFO: Preserving source tree in /home/sonny/openbmc/build/romulus/workspace/attic/sources/linux-aspeed.20251203163815
If you no longer need it then please delete it manually.
It is also possible to reuse it via devtool source tree argument.
```


測試結果對比

指令	修改前	修改後
i2cdetect -l	顯示 i2c-1 到 i2c-12	顯示 i2c-1 到 i2c-11 (無 i2c-12)
i2cdetect -y 12	正常顯示 I2C 地址掃描表	Error: No such file or directory

結論

本次 L3 作業成功完成以下目標：

項目	狀態	說明
分析硬體配置	✓ 完成	了解 AST2500 支援 14 個 I2C, romulus 啟用 12 個
修改 Kernel DTS	✓ 完成	將 i2c12 的 status 從 "okay" 改為 "disabled"
編譯與測試	✓ 完成	使用 devtool 工作流程, 成功驗證 i2c12 已停用
環境還原	✓ 完成	使用 devtool reset 還原乾淨環境

透過本次實作, 學習了以下技術：

- Device Tree Source (DTS) 的結構與修改方式
- I2C Bus、Controller、記憶體位址的關係
- devtool 開發工作流程 (modify → build → reset)
- i2cdetect 指令驗證 I2C 狀態

✓ L3 作業完成